The Odyssey of European Digital Sovereignty and a Knowledge Base to Orient Newcomers

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Abstract. European digital sovereignty lies at the centre of many legal, industrial, and technological initiatives launched by the European Commission (EC). Yet, information about these initiatives is scattered, and newcomers may struggle to grasp the bigger picture. This paper offers a primer to help newcomers understand the evolution of European digital sovereignty, discover and correlate key strategic initiatives via an open knowledge base, and recognize the challenges still ahead.

Keywords: European Digital Sovereignty · Open Knowledge Base · Dashboard

1 Introduction

Context. European digital sovereignty has become an important topic in recent years, as evident from the many initiatives launched by the European Commission (EC) since 2020 in legal, industrial, and technological policy areas [7,27]. European digital sovereignty can be defined as the ability of the European Union (EU) to act independently in the digital domain by developing and controlling core assets, infrastructures, and data, as well as carrying out related innovation activities to reduce technological dependencies — which may entail economic and security risks — while remaining open to global collaboration [6]. In other words, European digital sovereignty is a multifaceted topic that involves technical, operational, and governance aspects within the digital domain.

Motivation. Being a major driver for the digital domain, European digital sovereignty has a direct impact on industrial development and research, and recent EU funding programs explicitly aim to strengthen it.[4] At the same time, navigating the many related EC initiatives requires considerable time and effort, especially from newcomers, as information is scattered across various EU institutional and individual project websites. Moreover, EC initiatives often build on, complement, refer to, or amend one another, and readers may need to trace these connections by hand. European digital sovereignty has already been studied from specific viewpoints, like economy and geopolitics [3] — these are outside the scope of this work. However, to the best of our knowledge, a resource collecting and correlating (information on) EC initiatives does not exist. As a result,

[4] https://digital-strategy.ec.europa.eu/en/activities/digital-programme

industry practitioners and researchers may struggle to keep track of and cross-reference related EC initiatives and get a sense of how European digital sovereignty is unfolding.

Contributions. We provide a primer to orient newcomers in the *mare magnum* of EC initiatives for European digital sovereignty with the following contributions: (*i*) we discuss how European digital sovereignty has acquired importance and urgency in Section 2; (*ii*) we illustrate recent key strategic initiatives in Section 3; (*iii*) we propose a knowledge base (and a web dashboard) collecting information on EC initiatives in Section 4; (*iv*) we identify challenges still ahead and offer some further considerations in Section 5. We conclude the paper with final remarks and future work in Section 6.

2 From Awareness to Capability Building, in Brief

The debate on the technological dependencies of the EU on foreign actors — and their implications — has been ongoing since the late 20th century. To highlight the most visible shifts in the debate, we simplify the evolution of European digital sovereignty into three periods, each summarizing major events, EC activities, and "lessons learned" that shaped the debate and led to the current European posture on digital sovereignty.

Surveillance Controversies (2001–2015). Following earlier press reports, in 2001 the European Parliament (EP) presented serious reasons to believe in the existence of the Echelon surveillance program, a global telecommunications interception network operated by the "Five Eyes" intelligence alliance — United States (US), United Kingdom, Canada, Australia, and New Zealand [28]. The program was (at least partially) aimed at intercepting non-military private communications of European governments, organisations, and noteworthy individuals. Then, in June 2013, Edward Snowden disclosed classified documents on other mass surveillance programs of the US [21]. These documents revealed that US intelligence agencies have had continuous access to the personal data of European citizens stored by major US Cloud providers such as Google and Microsoft. The EP officially condemned this kind of programs [21], even though some Member States were even involved in them.[5,6] These controversies raised an initial awareness of the risks of digital communication, called for widespread adoption of open source cryptography software [28], and showed that technological dependencies — in particular, reliance on foreign Cloud providers — could (and did) expose European data to foreign countries without the knowledge (or meaningful consent) of those involved.

Legal Consolidation (2015–2020). Influenced by previous events, this period saw broader debates on data protection, international data transfers, and the limits of legal protection. Following the Snowden Disclosures, Maximilian Schrems challenged Facebook over transferring his data to the US.[7] The resulting Court of Justice of the

[5] https://www.lemonde.fr/technologies/article/2013/11/29/la-france-precieux-partenaire-de-l-espionnage-de-la-nsa_3522653_651865.html

[6] https://www.spiegel.de/international/world/german-intelligence-sends-massive-amounts-of-data-to-the-nsa-a-914821.html

[7] https://noyb.eu/en/project/eu-us-transfers

European Union (CJEU) rulings, known as Schrems I [4] (2015) and Schrems II [5] (2020), established that US laws did allow disproportionate access to European citizens' data by US authorities, invalidating the Safe Harbour and Privacy Shield EU-US data transfer agreements.[8] Around the same time, the General Data Protection Regulation (GDPR) [22] (adopted in April 2016, enforceable since May 2018) harmonized data protection across the EU and strengthened the rights of European citizens over their data. During this period, the EU sought to extend its data protection policies beyond its borders, and some foreign organizations even fully adopted them, and not just for European data — a phenomenon called *Brussels Effect* [2]. However, the EU also realized that legal instruments alone cannot prevent foreign laws from bypassing EU ones. Currently, the US CLOUD Act still "allows American authorities to compel US-based technology companies to provide requested data, regardless of where that data are stored globally" without notifying data owners;[9] this applies both to personal and business-sensitive data.

Capability Building (2020–Now). Since 2020, the EC has complemented legal activities (insufficient on their own[10]) with industrial and technological activities [23], with a notable intensification of methods across its two most recent mandates.[11] The first mandate pursued an *open strategic autonomy*, where the main objectives were reducing risks due to technological dependencies and improving supply chain resilience while remaining committed to open markets and global cooperation [10]. The second mandate, urged by the new political setting between the EU and US since 2025[12], pursues *digital sovereignty* with a stronger emphasis on control in the digital domain and the capability of enforcing (and not only making) EU choices based on European values, so as not to have to entrust them to foreign actors [27,3]. The two terms are strongly related but differ in perspective: open strategic autonomy is about reducing technological dependencies, while digital sovereignty is about exercising control [3], a more assertive evolution.

In conclusion, the three periods show three incremental stages in the building of EU's posture on digital sovereignty: awareness after Echelon and the Snowden Disclosures ("recognize the problem"), legal rights after Schrems and the GDPR ("solve the problem by regulating others"), and now the attempt to build industrial and technological capacity to truly exercise control in the digital domain ("solve the problem ourselves").

3 Key Strategic Initiatives for Capability Building

Since 2020, the EC has launched several initiatives to achieve a variety of objectives related to capability building in the digital domain, encompassing the reduction of

[8] A third agreement was proposed with the EU-US Data Privacy Framework (July 2023). Schrems's non-profit organization None Of Your Business (NOYB) announced it will challenge it — see https://www.jentis.com/blog/noyb-will-challenge-the-new-data-privacy-framework.

[9] https://www.theregister.com/off-prem/2025/12/22/europe-gets-serious-about-cutting-us-digital-umbilical-cord/2654348

[10] https://thenewstack.io/a-middle-path-for-data-sovereignty-bring-your-own-cloud/

[11] The first mandate of the EC led by President Ursula von der Leyen ran from 1st December 2019 to 30th November 2024; the second mandate started on 1st December 2024 and is ongoing.

[12] https://www.euractiv.com/news/europe-readies-digital-infrastructure-push-in-2026/

technological dependencies, the securing of supply chains, and the pursuit of open strategic autonomy and digital sovereignty. With "initiatives", we refer to *Regulations*, *Decisions*, and *Directives* (that are binding), *Communications* (that are non-binding) — in the context of EU legislation, these words have a specific meaning[13] — and bundles thereof (i.e., *Packages*, *Omnibuses*, and *Action Plans*). We divide these initiatives into three tiers: tier 1 comprises overarching digital strategies, tier 2 comprises initiatives specific to a technology area, and tier 3 comprises binding initiatives stemming from tiers 1 and 2. To keep the discussion on the (broad) subject of European digital sovereignty, we consider only initiatives satisfying three selection criteria: initiatives must be EU-level, cross-sector, and launched since 2020 (the "Capability Building" period in Section 2); we exclude sector-specific (e.g., energy, healthcare, space) and national-level initiatives.

Tier 1 Initiatives. The main reference for the EC work in the 2019–2024 mandate is the **Shaping Europe's Digital Future** *Communication* of February 2020 [10], which sets a vision on three pillars: digital technologies that benefit people, a fair and competitive market for EU organizations, and a truly open and democratic society respectful of European fundamental rights. Such a vision was translated into measurable targets for 2030 by the more operative "2030 Digital Compass: the European way for the Digital Decade" (or, simply, **Digital Compass**) *Communication* of March 2021 [11], referring to the decade 2020-2030. The Digital Compass was implemented by the Digital Decade Policy Programme 2030 (or, simply, **Digital Decade**) *Decision* of December 2022 [23]. Instead, the main reference for the EC work in the 2025–2030 mandate is the **Competitiveness Compass** *Communication* of January 2025 [14], which builds directly on the Draghi and Letta Reports [7,26] and, again, sets a vision on three pillars: close the innovation gap, reduce foreign dependencies, and decarbonize the EU digital economy. Finally, in November 2025 the EC has started a process to define a collection of amendments for simplifying and consolidating the set of those initiatives concerning the digital domain;[14] the process was initially envisioned in the **Digital *Package***[15] and now it is being operationalized with the **Digital Omnibus** *Regulation* proposal.[16]

Tier 2 Initiatives. Stemming from tier 1, tier 2 initiatives concern a specific technology area among the following: data, Artificial Intelligence (AI), semiconductors, Cloud and edge computing, cybersecurity and digital identity, and quantum computing.

[13] Regulations (sometimes called acts) are binding and applicable in all Member States uniformly without national implementation (transposition). Decisions are similarly binding and applicable but target specific addressees (e.g., an EU country). Directives are binding as to the result to be achieved by Member States, which can decide how to achieve it by transposing directives into national laws. Communications (or Joint Communications, when foreign or security policy is concerned) are non-binding documents setting out political priorities and a roadmap of regulatory, investment, and policy initiatives; the word strategy is often used in this context. Initiatives may be bundled for coherence and communication purposes into a package (when on a specific subject), an omnibus (when only amending existing EU laws), or an action plan (when concrete, operational, and focused on a specific aspect of a subject).

[14] This set is also referred to as "Digital Rulebook" or "Digital Acquis" (which are only informally defined, and the list of included initiatives is somewhat fluid and not a fixed canonical set).

[15] https://digital-strategy.ec.europa.eu/en/policies/digital-rulebook

[16] https://digital-strategy.ec.europa.eu/en/library/digital-omnibus-regulation-proposal

Data are a central asset of the EU digital economy. Stemming from **Shaping Europe's Digital Future** (and published the same day), the vision to protect and valorize data was first set in the **European Strategy for Data** *Communication* [9], proposing the so-called European market for data by facilitating data access, sharing, and usage via Common European Data Spaces, promoting data-driven innovation, and a fair allocation of the value that data produces amongst the participants in the data economy. In November 2025, stemming from the **Digital *Package*** and being operationalized via the **Digital Omnibus**, the new **Data Union Strategy** *Communication* updates the **European Strategy for Data** to increase the availability of data for AI development [15].

In this regard, the EC aims to realize the potential of AI while addressing challenges like underuse and overuse, liability, and threats to fundamental rights. Published with **Shaping Europe's Digital Future**, the EC "White paper on AI: a European approach to excellence and trust"[17] sets the initial vision to make the EC a leader for trustworthy and human-centric AI by combining investments, research, and innovation. The EC only recently set a broader strategy for the EU ambitions to become a "leading AI continent" with the **AI Continent Action Plan** *Communication* of April 2025 [12] (stemming from the **Competitiveness Compass**), followed by the **Apply AI Strategy** *Communication* of October 2025 which encourages an "AI-first" policy for organizations [13].

Concerning semiconductors, the EC has not published a dedicated strategy: related initiatives draw from the measurable targets defined in the **Digital Compass** and the joint declaration "A European Initiative on Processors and Semiconductor Technologies" signed by 22 Member States in December 2020[18] for open strategic autonomy.

A dedicated strategy is also missing for Cloud and edge computing. Still, we can mention two noteworthy initiatives: Gaia-X[19] — established in 2019 to build a federated and interoperable Cloud infrastructure for Europe — and the Important Project of Common European Interest on Cloud Infrastructure and Services (IPCEI-CIS)[20] — approved in December 2023 to support research, development, and industrial deployment of European innovations in Cloud and edge computing, enabling a Cloud to edge continuum across multiple providers through interoperable technologies and open source software.

The **EU's Cybersecurity Strategy for the Digital Decade** (joint) *Communication* of December 2020 [18] is the basis for initiatives on cybersecurity and digital identity. Stemming from **Shaping Europe's Digital Future** and the third cybersecurity strategy in a series, it sets the EC vision for regulatory, investment, and policy initiatives to protect essential services and withstand, respond to, and recover from cyberattacks.

Finally, stemming from the **Competitiveness Compass**, the EC published in July 2025 the **Quantum Europe Strategy** to close the EU gap in quantum technologies [16].

Tier 3 Initiatives. The EC has published many binding initiatives stemming from tiers 1 and 2, such as the **Data Governance *Act* (DGA)**, the **AI *Act***, and the **NIS 2 *Directive***. Due to space limitations, we do not list them here and refer the reader to Section 4.

[17] https://commission.europa.eu/publications/white-paper-artificial-intelligence-european-approach-excellence-and-trust_en

[18] https://digital-strategy.ec.europa.eu/en/library/joint-declaration-processors-and-semiconductor-technologies

[19] https://gaia-x.eu/

[20] https://www.ipcei-cis.eu/

4 An Open Knowledge Base for European Digital Sovereignty

As noted in Section 1, European digital sovereignty is pursued not through a single initiative but via a continually evolving and interconnected set of strategies, regulations, directives, funding programs and industrial activities. For instance, to understand the relevance of the **European Strategy for Data**, one should know that it was later updated by the **Data Union Strategy** to meet the data needs of AI development (Section 3). Even though this information is publicly available, its fragmentation[21] and (to the best of our knowledge) the lack of a resource collecting and contextualizing it entail considerable time and cognitive costs, especially for newcomers. Motivated by this concern, we build an open knowledge base that collects the EC initiatives across all three tiers described in Section 3 made entirely from information available on the EU institutional website for the digital domain,[22] the Official Journal of the EU,[23] and individual — but still official — project sites. We manually process these sources to identify relevant initiatives and the explicit connections among them, and structure the result as a `.yaml` file.[24] At the time of writing, the knowledge base includes 44 initiatives along with their type, lifecycle, technology area (see tier 2 in Section 3), and relationships with other initiatives (e.g., "builds on", "amends", "replaces"). We aim to keep the knowledge base updated to always reflect the state of the art of EC initiatives on European digital sovereignty.

The European Digital Sovereignty Atlas. As a first application of the knowledge base, we develop the **European Digital Sovereignty Atlas** (or, simply, the Atlas) web dashboard.[25] Via interactive charts (see Figure 1), the Atlas shows the lifecycle of initiatives, groups them by technology area and digital strategy (tiers 1 and 2) using the connections in the knowledge base, and displays various statistics. Moreover, the Atlas can act as a point of access for Large Language Model (LLM)-based agents, which can download the knowledge base (.yaml) as an up-to-date source for Retrieval-Augmented Generation (RAG) pipelines and other automated retrieval methods. This possibility opens several directions for future work, such as validating the accuracy of an agent's output against the Atlas as ground truth, or comparing it to a generic web search to assess how much the Atlas improves the accuracy, verbosity, and completeness of the agent's output.

5 Considerations and Challenges Still Ahead

Despite the many initiatives for European digital sovereignty launched by the EC and recounted in the knowledge base and the Atlas, visible results may not yet fully match this ambition, and some ethical and political tensions may persist.

Technological Dependencies are Still Predominant. In 2023, 80% of all digital products, services, and infrastructures used in the EU came from foreign countries [7]. US Cloud

[21] https://sphericalcowconsulting.com/2026/09/01/standards-transparency/
[22] https://digital-strategy.ec.europa.eu/
[23] https://eur-lex.europa.eu/homepage.html
[24] https://github.com/aleph-fbk/the-odyssey-of-european-digital-sovereignty
[25] https://aleph.fbk.eu/the-odyssey-of-european-digital-sovereignty/atlas/

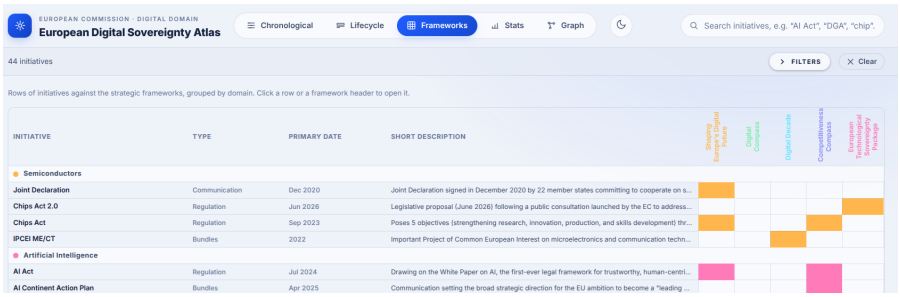


Fig. 1. The Frameworks view of the European Digital Sovereignty Atlas.

providers held 70% of the European Cloud market at the end of 2025, while the share of European Cloud providers dropped from 29% in 2017 to 15% in 2025.[26] Notably, 77.53% of EU organizations using Cloud services (corresponding to 40.89% of all EU organizations) were classified as highly dependent on such Cloud services.[27] In other words, the operations of around 35% of EU organizations hinge on the availability of foreign Cloud providers. Similarly, the EC expected the EU to achieve the target of 20% of the global semiconductor production value by 2030.[28] However, in April 2025 the European Court of Auditors found this target "very unlikely" to be met, as the EC's own forecast shows the European share rising only from 9.8% in 2022 to 11.7% by 2030 — meeting the original 20% target would have required roughly quadrupling production capacity [20]. According to the Draghi report, closing this broader technological gap would require an additional €750–800 billion in investment per year [7].

Fragmented Governance and Understanding of Digital Sovereignty. Fragmentation across 27 Member States remains an obstacle to the effective implementation of EC initiatives. As an example, Gaia-X has been criticized for a lack of clear governance, alongside bureaucratic bloat, internal disagreements, and the involvement of foreign Cloud providers undermining the "sovereignty" premise, resulting in unclear progress toward anything concrete.[29,30] Partly, this was because the definition and implications of "digital sovereignty" are still not agreed upon by all European institutions, Member States, and organizations [3]. In Gaia-X specifically, the term "digital sovereignty" takes six different meanings across security, economic, and rights policy areas [1]. Therefore, the success of EC initiatives will likely depend less on new laws — where the EU has a working track record, as shown by the Brussels Effect (Section 2) — and more on effective governance and clearer goals. Luckily, as shown below, work toward a common (and measurable) understanding of European digital sovereignty is underway [6].

[26] https://www.srgresearch.com/articles/european-cloud-providers-local-market-share-now-holds-steady-at-15

[27] https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Cloud_computing_-_statistics_on_the_use_by_enterprises

[28] https://digital-strategy.ec.europa.eu/en/policies/european-chips-act

[29] https://www.politico.eu/article/chaos-and-infighting-are-killing-europes-grand-cloud-project/

[30] https://www.politico.eu/article/anatomy-franco-german-tech-misfire-american-dependence/

Scores, Risk Levels, and Certifications. While European digital sovereignty was previously more of a political claim subject to interpretation, EC initiatives under the **Digital Decade** now introduce scores, risk levels, and certifications that make it more easily measurable (either quantitatively or qualitatively). For instance, the draft of the **Cloud Sovereignty Framework** proposes a digital sovereignty score based on 5 Sovereignty Effectiveness Assurance Levels (SEALs) and 48 criteria [19], the **AI Act** defines 4 levels of risk for AI systems [24], and the **European Cybersecurity Certification Framework (ECCF)** introduces certification schemes for digital products and services [17].

From Data Access to Data Control. The shift from open strategic autonomy to a more assertive digital sovereignty is especially visible in EC initiatives on data: those from 2020–2024, such as the **DGA** and the Data Act, focused on removing barriers to data flow across the EU by promoting data reuse, voluntary data sharing, and availability through sectoral Common European Data Spaces. Initiatives from 2025 onward, such as the **Competitiveness Compass** and the **Technological Sovereignty *Package***, focus instead on data control by promoting a preference for European and open source solutions, data sovereignty, and capabilities building in Cloud computing and AI.

The Ethical Tension Between Privacy and Surveillance. Despite legal progress, the tension between privacy and state surveillance persists. Even after Schrems I and Schrems II, the problem of international data transfers remains open with the US CLOUD Act that poses an "irreconcilable legal conflict" with EU laws.[31] Internally, the EC is removing barriers to encourage the collection, sharing, and reuse of data to benefit competitiveness. However, making data useful at scale may come at the expense of data privacy, a conflict that requires careful negotiation. For instance, the **European Health Data Space *Regulation* (EHDS)** [25] defines primary uses of health data (which stay under the patient's control) but also secondary uses under conditions not requiring patients' consent. A similar reduction of data protection by design is included in some GDPR amendments being negotiated in the **Digital Omnibus**, like that narrowing the definition of personal data (dropped by the EC ensuing protests)[32] and that making the processing of personal data for AI automatically lawful.[33] Caution regarding state surveillance is particularly relevant in light of past experience within the EU, including the involvement of some Member States in the surveillance programs exposed by Edward Snowden and the ongoing debate on the **Chat Control** proposal threatening end-to-end encryption [8]. This points to a further risk: efforts to protect European data from foreign surveillance should not end up replacing it with surveillance by European authorities instead.

6 Conclusion

European digital sovereignty evolved from a simple political claim to a complex set of interconnected initiatives. As an entry point allowing newcomers to participate to

[31] https://www.theregister.com/off-prem/2025/12/22/europe-gets-serious-about-cutting-us-digital-umbilical-cord/2654348

[32] https://noyb.eu/en/eu-commission-about-wreck-core-principles-gdpr

[33] https://noyb.eu/en/ai-eu-member-states-plan-digital-expropriation-europeans-interest-ai-companies

the debate, this primer showed how the European posture on digital sovereignty came to be, surveyed key strategic initiatives, identified points of attention, and proposed a knowledge base on EC initiatives available via the **European Digital Sovereignty Atlas**.

Future Work. Besides LLM-based retrieval, we plan to assess accuracy via independent expert re-annotation of a sample of initiatives and connections (measuring agreement) and automated consistency checks, and usability via a user study using the System Usability Scale. We plan to keep the knowledge base updated, open to contributions, and extend its coverage to quantitative data (e.g., funding volumes) and national initiatives.

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References

1. Adler-Nissen, R., Eggeling, K.A.: The discursive struggle for digital sovereignty: Security, economy, rights and the cloud project Gaia-X. JCMS: Journal of Common Market Studies **62**, 993–1011 (2024). https://doi.org/10.1111/jcms.13594
2. Bradford, A.: The Brussels Effect: How the European Union Rules the World. Oxford University Press (2020). https://doi.org/10.1093/oso/9780190088583.001.0001
3. Broeders, D., Cristiano, F., Kaminska, M.: In search of digital sovereignty and strategic autonomy: Normative power Europe to the test of its geopolitical ambitions. JCMS: Journal of Common Market Studies **61**, 1261–1280 (2023). https://doi.org/10.1111/jcms.13462
4. Court of Justice of the European Union: Maximilian schrems v data protection commissioner (2015), https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:62014CJ0362
5. Court of Justice of the European Union: Data protection commissioner v facebook ireland limited and maximilian schrems (2020), https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:62018CJ0311
6. Di Marco, D., Thabit Gonzalez, S., Kotsev, A., Friis-Christensen, A., Minghini, M., King, M.: Open but not powerless: Towards a common understanding of EU digital sovereignty. Tech. Rep. JRC144908, European Commission, Joint Research Centre (2025), https://publications.jrc.ec.europa.eu/repository/handle/JRC144908
7. Draghi, M.: The future of European competitiveness (Sep 2024), https://commission.europa.eu/topics/competitiveness/draghi-report_en
8. EDPB, EDPS: EDPB-EDPS Joint Opinion 04/2022 on the Proposal for a Regulation of the European Parliament and of the Council laying down rules to prevent and combat child sexual abuse (Jul 2022), https://www.edpb.europa.eu/documents/legislative-opinion/edpb-edps-joint-opinion-042022-on-the-proposal-for-a-regulation-of_en
9. European Commission: A European strategy for data (Feb 2020), https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:52020DC0066
10. European Commission: Shaping Europe's digital future. Tech. Rep. COM(2020) 67 final, European Commission (Feb 2020), https://commission.europa.eu/system/files/2020-02/communication-shaping-europes-digital-future-feb2020_en_4.pdf

11. European Commission: 2030 digital compass — the european way for the digital decade (Mar 2021), https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:52021DC0118
12. European Commission: AI continent action plan (Apr 2025), https://digital-strategy.ec.europa.eu/en/library/ai-continent-action-plan
13. European Commission: Apply AI strategy (Oct 2025), https://digital-strategy.ec.europa.eu/en/policies/apply-ai
14. European Commission: A competitiveness compass for the EU. Tech. Rep. COM(2025) 30 final, European Commission (Jan 2025), https://commission.europa.eu/topics/competitiveness/competitiveness-compass_en
15. European Commission: Data union strategy: Unlocking data for AI (Nov 2025), https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:52025DC0835
16. European Commission: Quantum Europe Strategy, quantum Europe in a changing world (Jul 2025), https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:52025DC0363
17. European Commission: European cybersecurity certification framework (2026), https://digital-strategy.ec.europa.eu/en/policies/cybersecurity-certification-framework
18. European Commission, High Representative of the Union for Foreign Affairs and Security Policy: The EU's cybersecurity strategy for the digital decade (Dec 2020), https://digital-strategy.ec.europa.eu/en/policies/cybersecurity-strategy
19. European Commission, Directorate-General for Digital Services: Cloud sovereignty framework. Tech. rep., European Commission (Oct 2025), https://commission.europa.eu/document/download/09579818-64a6-4dd5-9577-446ab6219113_en
20. European Court of Auditors: Special Report 12/2025: The EU's strategy for microchips. Tech. rep., European Court of Auditors (Apr 2025), https://www.eca.europa.eu/en/publications/SR-2025-12
21. European Parliament: European Parliament resolution of 12 March 2014 on the US NSA surveillance programme, surveillance bodies in various Member States and their impact on EU citizens' fundamental rights and on transatlantic cooperation in justice and home affairs (2013/2188(INI)) (Mar 2014), https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52014IP0230, 2013/2188(INI)
22. European Parliament and Council of the European Union: Regulation (eu) 2016/679 (general data protection regulation) (Apr 2016), https://eur-lex.europa.eu/eli/reg/2016/679/oj/eng, oJ L 119, pp. 1–88
23. European Parliament and Council of the European Union: Decision (EU) 2022/2481 establishing the digital decade policy programme 2030 (Dec 2022), https://eur-lex.europa.eu/eli/dec/2022/2481/oj/eng, oJ L 323, 19.12.2022, p. 4–26
24. European Parliament and Council of the European Union: Regulation (eu) 2024/1689 (artificial intelligence act) (Jul 2024), https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=OJ:L_202401689
25. European Parliament and Council of the European Union: Regulation (eu) 2025/327 on the european health data space (Feb 2025), https://eur-lex.europa.eu/eli/reg/2025/327/oj/eng
26. Letta, E.: Much more than a market: Speed, security, solidarity. Tech. rep., European Council (Apr 2024), https://www.consilium.europa.eu/media/ny3j24sm/much-more-than-a-market-report-by-enrico-letta.pdf
27. von der Leyen, U.: Europe's choice: Political guidelines for the next European Commission 2024–2029. Tech. rep., European Commission (Jul 2024), https://commission.europa.eu/document/download/e6cd4328-673c-4e7a-8683-f63ffb2cf648_en
28. Piodi, F., Mombelli, I.: The ECHELON affair: The European Parliament and the global interception system. Tech. Rep. PE 538.877, European Parliament (Nov 2014). https://doi.org/10.2861/76176